

# ABG Motors Market Entry Analysis Report

Market Entry Analysis for ABG Motors in India

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## 1. Business Understanding

ABG Motors is a Japanese automobile manufacturer planning to expand into the Indian market. The objective of this project is to analyze customer data from Japan, build a classification model to predict customer purchases, and apply the model to the Indian dataset to estimate potential buyers. The company aims to identify at least **12,000 potential customers** before entering the Indian market.

## 2. Data Preprocessing

- Loaded the Japanese and Indian datasets using Pandas.
- Converted the maintenance date (DT\_MAINT) into datetime format.
- Created a new feature called **AGE\_CAR**.
- Converted Gender into numerical values for model training.
- Selected relevant features:
  - Current Age
  - Gender
  - Annual Income
  - Age of Car
- Standardized the features using **StandardScaler**.

## 3. Model Building

A **Logistic Regression** model was built because the target variable (Purchase) is binary (0 = No, 1 = Yes). The Japanese dataset was divided into **80% training data** and **20% testing data** to train and evaluate the model.

## 4. Model Performance

### Classification Report:

Accuracy : 68%

Precision : 0.68

Recall : 0.68

F1-Score : 0.68

The model achieved an accuracy of **68%**, indicating that it performs reasonably well in predicting customer purchases.

## 5. Interpretation of Model Coefficients

| Feature       | Coefficient | Interpretation                                                       |
|---------------|-------------|----------------------------------------------------------------------|
| Age of Car    | 0.846       | Customers with older cars are more likely to purchase a new vehicle. |
| Annual Income | 0.404       | Higher income increases the probability of purchasing.               |
| Gender        | 0.109       | Gender has a small positive influence on purchase decisions.         |
| Current Age   | -0.126      | Older customers are slightly less likely to purchase.                |

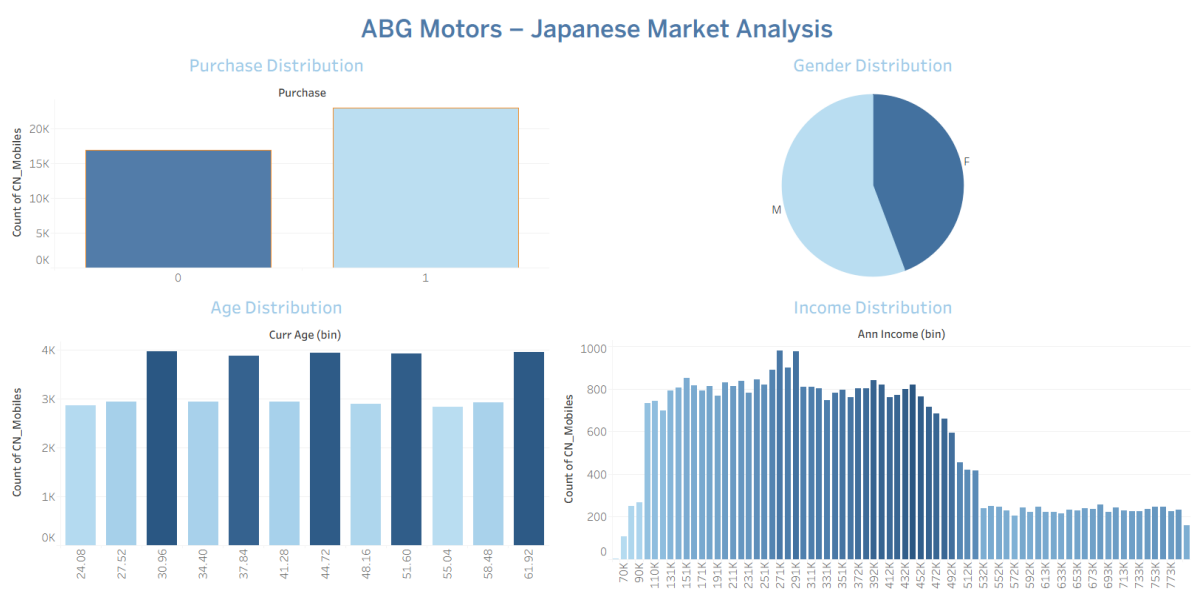
## 6. Prediction on Indian Dataset

The trained Logistic Regression model was applied to the Indian dataset.

**Predicted Potential Buyers: 67,231**

The predicted number of potential buyers (**67,231**) is significantly higher than the required target of **12,000**. This indicates that ABG Motors has strong market potential in India.

## 7. Tableau Dashboard



### Dashboard Insights

- Purchase distribution was analyzed.
- Gender distribution was visualized.
- Customer age distribution was examined.
- Annual income distribution was analyzed.
- Purchase behavior by gender was observed.

## 8. Conclusion

Based on the Logistic Regression model and Tableau analysis, ABG Motors has a promising opportunity to enter the Indian market. The model predicted **67,231 potential buyers**, which exceeds the required target of **12,000**. Therefore, the analysis suggests that expanding into the Indian market is likely to be profitable.